

Stretching and deforming

AS WELL AS MOVING OBJECTS FROM PLACE TO PLACE, FORCES CAN ALSO MAKE OBJECTS CHANGE SHAPE.

SEE ALSO

◀ 96–97 Properties of materials

◀ 98–99 States of matter

◀ 163 Plastics

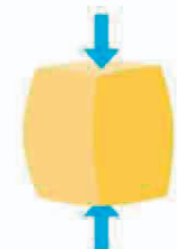
◀ 172–173 Forces and mass

When a force acts on an object that cannot move, or when a number of different forces act in different directions, they make the object's molecules (or other small parts of it) move closer together or further apart, so the whole object changes shape.

Types of distortion

The type of distortion an object undergoes depends on the number, directions, and strengths of forces acting on it, and also on its structure and composition. Many objects simply snap or shatter when strong forces act on them. Those that do not are referred to as deformable, such as modeling clay.

Graphene is one of the **strongest** and **most elastic** materials. It is made of sheets of carbon atoms that are connected together in hexagons.



△ Compression

When two or more forces act in opposite directions and meet at the same point inside an object, the object will compress and bulge out on all sides.



△ Tension

When two or more forces act in opposite directions and pull away from a object, they apply tension, and elastic objects will stretch in response.



△ Bending

When several forces act on an object in different places, the object will either snap (if it is brittle) or bend (if it is malleable). Many materials, like wood, bend a little, and then snap.



△ Torsion

Turning forces, or torques, that act in opposite directions, but affect different parts of an object, result in the object being twisted.



△ Shearing

When forces act in opposite directions at the ends of an object that is not free to spin, its ends will move in two different directions.

Deformation

Forces that change the shape of an object are known as stresses. The change in shape of the object in response is called a strain. When an object is stressed, three things may happen: it may break; it may change shape permanently (in which case it is said to be "plastic"); or it may change shape until the stress is removed—and then return to its original shape (elastic).

▷ Stress–strain curve

Many materials behave elastically for small distortions, and then will start to behave plastically. Finally, they will break. The forces required vary a lot, depending on the material.

